

Uses of React Native

React Native is an open-source framework for building Android and iOS applications. You can use JavaScript to access your app platform's APIs as well as to describe the UI using React components.

Native Components

Native components are platform-backed components. These components are invoked with JavaScript using React components. At runtime, React Native creates the corresponding Android and iOS views.

React Native and Expo

In React Native, the provided framework requires a basic knowledge of Xcode and Android Studio to build apps. Expo is a framework of tools and services for React Native that focuses on letting you build apps without Xcode and Android Studio.

JS and UI Threads

The *JS thread* executes the bundled JavaScript code, which then sends actions or requests to the *UI thread*.

Core vs. Native Components

Each *core component* is implemented using its respective *native component* counterpart. Because *native components* are platform-specific, they could look different when rendered on different platforms.

Entry Point

All functionality of the React Native app must be included in a single React component, exported as default from the *entry point* file.

```
import * as React from 'react';

const App = () => (
  // All functionality goes here
);

export default App;
```

Packages

You can import different packages to gain access to different native APIs or functionality.

```
import { Camera } from 'expo-camera';
import { MapView } from 'react-native-maps';
```

Expo SDK

The Expo SDK is a collection of packages that allows access to some of the most-used APIs for all apps.

Expo CLI

Expo provides a CLI tool as the primary interface for developers to start writing code for a project.

```
$ expo init
```

Expo Go

“Expo Go” is a development client for Android and iOS that can load the JS part of the project. This allows you to preview your app in development.

Expo Snack

“Expo Snack” is a web-based playground that allows you to write and run React Native apps in the browser.

Benefits and Drawbacks

Expo and React Native are ideal tools when you need (i) to run an app on multiple platforms, (ii) direct access to native functionality, and/or (iii) only have basic web development and native platform understanding.

Expo and React Native aren't ideal tools when you need (i) absolute performance, (ii) cutting edge features which are just released by the platform, and/or (iii) complex and big applications.

Native Code

“Native code” are instructions for how to operate the React Native framework on each platform, such as iOS and Android.



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